Lab 4: Networking in Cloud

A **load balancer** is a critical component in modern networking and cloud infrastructure that helps distribute incoming network traffic across multiple servers or resources. Its primary goal is to ensure that no single server or resource is overwhelmed with too much traffic, thus improving the **availability, reliability, and scalability** of applications or services. Load balancers play an essential role in networking by optimizing traffic flow, ensuring fault tolerance, and improving overall application performance.

Load balancers operate at the **network layer** of the OSI model (Layer 4) or the **application layer** (Layer 7), depending on the type of load balancer. Their job is to distribute incoming client requests to multiple backend servers (also known as "pool" or "targets"), ensuring that no single server becomes a bottleneck.

- **Layer 4 Load Balancer (Network Layer)**: Routes traffic based on **IP address** and **port number**.
- **Layer 7 Load Balancer (Application Layer)**: Routes traffic based on more advanced criteria, such as **HTTP headers**, **cookies**, **URL paths**, etc.

Objective:

- Implement a Load Balancer to distribute traffic across 2 EC2 instances. Test by accessing the Load Balancer's URL and verifying that the traffic is routed correctly to the instances.

Submission:

- In your lab setup, what would happen if one of the backend servers in your load balancer pool becomes unavailable (e.g., crashes)?
- Why is load balancing important for highly available web applications?

  https://www.youtube.com/watch?v=KNr3Kq7cah8


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